1)
The origin, $(0, 0)$

2) a)
$\begin{pmatrix} 1 & 0 \\ 5 & -1 \end{pmatrix}\begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$
$\therefore (2, 5)$ is invariant under the transformation

2) b)
$\begin{pmatrix} 6 & -2 \\ 10 & -3 \end{pmatrix}\begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$
$\therefore (2, 5)$ is invariant under the transformation

2) c)
$\begin{pmatrix} 15 & -5 \\ 11 & -4 \end{pmatrix}\begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$
$\therefore (2, 5)$ is not invariant under the transformation

2) d)
$\begin{pmatrix} 21 & -8 \\ 100 & -39 \end{pmatrix}\begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$
$\therefore (2, 5)$ is invariant under the transformation

2) e)
$\begin{pmatrix} 50 & -21 \\ 31 & -12 \end{pmatrix}\begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$
$\therefore (2, 5)$ is not invariant under the transformation

3) a)
$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$
$\therefore (1, -2, -1)$ is invariant under the transformation

3) b)
$\begin{pmatrix} 2 & -1 & 3 \\ 3 & 4 & -3 \\ 2 & -1 & 5 \end{pmatrix}\begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$
$\therefore (1, -2, -1)$ is invariant under the transformation

3) c)

$$\begin{pmatrix} 19 & 1 & 16 \\ 12 & 11 & -8 \\ -5 & 3 & -10 \end{pmatrix}\begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

$\therefore (1, -2, -1)$ is invariant under the transformation

3) d)

$$\begin{pmatrix} \frac{1}{2} & -\frac{5}{12} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{4} & \frac{5}{6} \\ \frac{3}{2} & \frac{6}{5} & \frac{1}{10} \end{pmatrix}\begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{5}{3} \\ -1 \end{pmatrix}$$

$\therefore (1, -2, -1)$ is not invariant under the transformation

4) a)

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} y \\ -x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$y = x$

$-x = y$

The only solution to $y = x$ and $y = -x$ is $x = 0$ and $y = 0$

4) b)

$$\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{\sqrt{3}}{2}x - \frac{1}{2}y \\ \frac{1}{2}x + \frac{\sqrt{3}}{2}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$\frac{\sqrt{3}}{2}x - \frac{1}{2}y = x \Rightarrow \sqrt{3}x - y = 2x \Rightarrow y = (\sqrt{3} - 2)x$

$\frac{1}{2}x + \frac{\sqrt{3}}{2}y = y \Rightarrow x + \sqrt{3}y = y \Rightarrow x = y - \sqrt{3}y \Rightarrow y = \frac{1}{1-\sqrt{3}}x \Rightarrow y = \frac{-1-\sqrt{3}}{2}x$

The only solution to $y = (\sqrt{3} - 2)x$ and $y = \frac{-1-\sqrt{3}}{2}x$ is $x = 0$ and $y = 0$

4) c)

$$\begin{pmatrix} -1 & 5 \\ -3 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -x + 5y \\ -3x + 2y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-x + 5y = x \Rightarrow 2x - 5y = 0 \Rightarrow y = \frac{2}{5}x$

$-3x + 2y = y \Rightarrow 3x - y = 0 \Rightarrow y = 3x$

The only solution to $y = \frac{2}{5}x$ and $y = 3x$ is $x = 0$ and $y = 0$

4) d)

$$\begin{pmatrix}1 & 1 & 0\\ 0 & 1 & 1\\ 1 & 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}x\\ y\\ z\end{pmatrix}\Rightarrow\begin{pmatrix}x+y\\ y+z\\ x+z\end{pmatrix}=\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$x+y=x\Rightarrow y=0$ (1)

$y+z=y\Rightarrow z=0$ (2)

$x+z=z\Rightarrow x=0$ (3)

The only solution is $x=0,\ y=0$ and $z=0$

4) e)

$$\begin{pmatrix}-1 & 1 & 2\\ -1 & 2 & 0\\ 0 & 3 & -2\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}x\\ y\\ z\end{pmatrix}\Rightarrow\begin{pmatrix}-x+y+2z\\ -x+2y\\ 3y-2z\end{pmatrix}=\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$-x+y+2z=x\Rightarrow -2x+y+2z=0$ (1)

$-x+2y=y\Rightarrow -x+y=0$ (2)

$3y-2z=z\Rightarrow 3y-3z=0\Rightarrow y-z=0$ (3)

The only solution is $x=0,\ y=0$ and $z=0$

4) f)

$$\begin{pmatrix}2 & 3 & -1\\ -3 & 1 & 2\\ -2 & 2 & 3\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}x\\ y\\ z\end{pmatrix}\Rightarrow\begin{pmatrix}2x+3y-z\\ -3x+y+2z\\ -2x+2y+3z\end{pmatrix}=\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$2x+3y-z=x\Rightarrow x+3y-z=0$ (1)

$-3x+y+2z=y\Rightarrow -3x+2z=0$ (2)

$-2x+2y+3z=z\Rightarrow -2x+2y+2z=0\Rightarrow -x+y+z=0$ (3)

The only solution is $x=0,\ y=0$ and $z=0$

5)

$$\begin{pmatrix}9 & 2\sqrt{5}\\ 16 & 1+4\sqrt{5}\end{pmatrix}\begin{pmatrix}\sqrt{5}\\ -4\end{pmatrix}=\begin{pmatrix}9\sqrt{5}-8\sqrt{5}\\ 16\sqrt{5}-4\left(1+4\sqrt{5}\right)\end{pmatrix}=\begin{pmatrix}\sqrt{5}\\ 16\sqrt{5}-4-16\sqrt{5}\end{pmatrix}=\begin{pmatrix}\sqrt{5}\\ -4\end{pmatrix}$$

$\therefore\left(\sqrt{5},-4\right)$ is an invariant point

6)

$$\begin{pmatrix} \frac{3}{11} & \frac{10}{11} \\ \frac{3}{5} & \frac{1}{4} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{3}{11}x + \frac{10}{11}y \\ \frac{3}{5}x + \frac{1}{4}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$\frac{3}{11}x + \frac{10}{11}y = x \Rightarrow -\frac{8}{11}x + \frac{10}{11}y = 0 \Rightarrow -8x + 10y = 0 \Rightarrow y = \frac{8}{10}x \Rightarrow y = \frac{4}{5}x$

$\frac{3}{5}x + \frac{1}{4}y = y \Rightarrow \frac{3}{5}x - \frac{3}{4}y = 0 \Rightarrow 12x - 15y = 0 \Rightarrow y = \frac{12}{15}x \Rightarrow y = \frac{4}{5}x$

So another point that would be invariant is $\left(1, \frac{4}{5}\right)$

7)

$$\begin{pmatrix} -\frac{1}{8} & -\frac{27}{16} \\ -\frac{1}{4} & \frac{5}{8} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -\frac{1}{8}x - \frac{27}{16}y \\ -\frac{1}{4}x + \frac{5}{8}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-\frac{1}{8}x - \frac{27}{16}y = x \Rightarrow \frac{9}{8}x + \frac{27}{16}y = 0 \Rightarrow 18x + 27y = 0 \Rightarrow y = -\frac{18}{27}x \Rightarrow y = -\frac{2}{3}x$

$-\frac{1}{4}x + \frac{5}{8}y = y \Rightarrow \frac{1}{4}x + \frac{3}{8}y = 0 \Rightarrow 2x + 3y = 0 \Rightarrow y = -\frac{2}{3}x$

So another point that would be invariant is $\left(1, -\frac{2}{3}\right)$

8)

$$\begin{pmatrix} \frac{83}{19} & -\frac{24}{19} \\ -\frac{32}{19} & \frac{31}{19} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{83}{19}x - \frac{24}{19}y \\ -\frac{32}{19}x + \frac{31}{19}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$\frac{83}{19}x - \frac{24}{19}y = x \Rightarrow \frac{64}{19}x - \frac{24}{19}y = 0 \Rightarrow 64x - 24y = 0 \Rightarrow y = \frac{64}{24}x \Rightarrow y = \frac{8}{3}x$

$-\frac{32}{19}x + \frac{31}{19}y = y \Rightarrow -\frac{32}{19}x + \frac{12}{19}y = 0 \Rightarrow -32x + 12y = 0 \Rightarrow y = \frac{32}{12}x \Rightarrow y = \frac{8}{3}x$

So two points that satisfy both of these equations are $(0, 0)$ and $\left(1, \frac{8}{3}\right)$

9)

$$\begin{pmatrix} 2 & 0 & 2 \\ 1 & 3 & 0 \\ 0 & 2 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} 2x + 2z \\ x + 3y \\ 2y - z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$2x + 2z = x \Rightarrow x + 2z = 0 \Rightarrow x = -2z$ (1)

$x + 3y = y \Rightarrow x + 2y = 0 \Rightarrow x = -2y$ (2)

$2y - z = z \Rightarrow 2y - 2z = 0 \Rightarrow y = z$ (3)

Let $z = 1$ then $x = -2$ and $y = 1$

So another point that would be invariant is $(-2, 1, 1)$

10)

$$\begin{pmatrix} 1 & -4 & -6 \\ 4 & -3 & 2 \\ -3 & 2 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} x-4y-6z \\ 4x-3y+2z \\ -3x+2y-2z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$x - 4y - 6z = x \Rightarrow -4y - 6z = 0 \Rightarrow 2y + 3z = 0$ (1)

$4x - 3y + 2z = y \Rightarrow 4x - 4y + 2z = 0 \Rightarrow 2x - 2y + z = 0$ (2)

$-3x + 2y - 2z = z \Rightarrow -3x + 2y - 3z = 0$ (3)

Rearrange (1): $y = -\frac{3}{2}z$ and substitute into (2):

$2x - 2\left(-\frac{3}{2}z\right) + z = 0 \Rightarrow 2x + 4z = 0 \Rightarrow x + 2z = 0 \Rightarrow x = -2z$ (4)

Let $z = 2$ then $x = -4$ and $y = -3$

So another point that would be invariant is $(-4, -3, 2)$

11)

$$\begin{pmatrix} \frac{10}{m+3} & \frac{m-7}{6m+18} \\ \frac{m+1}{m+3} & \frac{5m+17}{6m+18} \end{pmatrix}\begin{pmatrix} 1 \\ 6 \end{pmatrix} = \begin{pmatrix} \frac{10}{m+3} + \frac{6(m-7)}{6m+18} \\ \frac{m+1}{m+3} + \frac{6(5m+17)}{6m+18} \end{pmatrix} = \begin{pmatrix} \frac{10}{m+3} + \frac{m-7}{m+3} \\ \frac{m+1}{m+3} + \frac{5m+17}{m+3} \end{pmatrix} = \begin{pmatrix} \frac{10+m-7}{m+3} \\ \frac{m+1+5m+17}{m+3} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{m+3}{m+3} \\ \frac{6m+18}{m+3} \end{pmatrix} = \begin{pmatrix} \frac{m+3}{m+3} \\ \frac{6(m+3)}{m+3} \end{pmatrix} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$$

$\therefore (1, 6)$ is invariant under the transformation

12) a)

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -x \\ -y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-x = x \Rightarrow x = 0$

$-y = y \Rightarrow y = 0$

So $(0, 0)$ is the only invariant point.

This is because the matrix represents a rotation by 180° about the origin.

12) b)

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$y = x$

$x = y \Rightarrow y = x$

So $y = x$ is a line of invariant points.

This is because the matrix represents a reflection in the line $y = x$

12) c)

$\begin{pmatrix}4 & 0\\ 0 & 4\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}\Rightarrow\begin{pmatrix}4x\\ 4y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}$

$4x=x\Rightarrow x=0$

$4y=y\Rightarrow y=0$

So $(0,0)$ is the only invariant point.

This is because the matrix represents an enlargement about the origin by scale factor 4

12) d)

$\begin{pmatrix}0 & -1\\ -1 & 0\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}\Rightarrow\begin{pmatrix}-y\\ -x\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}$

$-y=x\Rightarrow y=-x$

$-x=y\Rightarrow y=-x$

So $y=-x$ is a line of invariant points.

This is because the matrix represents a reflection in the line $y=-x$

12) e)

$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}\Rightarrow\begin{pmatrix}-y\\ x\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}$

$-y=x\Rightarrow y=-x$

$x=y\Rightarrow y=x$

So $(0,0)$ is the only invariant point.

This is because the matrix represents a rotation by $90°$, anticlockwise about the origin.

12) f)

$\begin{pmatrix}1 & 6\\ 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}\Rightarrow\begin{pmatrix}x+6y\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}$

$x+6y=x\Rightarrow 6y=0\Rightarrow y=0$

$y=y$

So $y=0$ is a line of invariant points.

This is because the matrix represents a shear, with the x-axis fixed, with $(0,1)$ mapped to $(6,1)$

12) g)

$\begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ -4x + y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$x = x$

$-4x + y = y \Rightarrow -4x = 0 \Rightarrow x = 0$

So $x = 0$ is a line of invariant points.

This is because the matrix represents a shear, with the y-axis fixed, with $(1, 0)$ mapped to $(1, -4)$

12) h)

$\begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} 4x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$4x = x \Rightarrow x = 0$

$y = y$

So $x = 0$ is a line of invariant points.

This is because the matrix represents a stretch, parallel to the x-axis, scale factor 4

12) i)

$\begin{pmatrix} -2 & -2 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -2x - 2y \\ 3x + 3y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$-2x - 2y = x \Rightarrow 3x + 2y = 0 \Rightarrow y = -\frac{3}{2}x$

$3x + 3y = y \Rightarrow 3x + 2y = 0 \Rightarrow y = -\frac{3}{2}x$

So $y = -\frac{3}{2}x$ is a line of invariant points.

12) j)

$\begin{pmatrix} -3 & -3 \\ 2 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -3x - 3y \\ 2x + 2y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$-3x - 3y = x \Rightarrow 4x + 3y = 0 \Rightarrow y = -\frac{4}{3}x$

$2x + 2y = y \Rightarrow 2x + y = 0 \Rightarrow y = -2x$

So $(0, 0)$ is the only invariant point.

13)

$\begin{pmatrix} 8 & 4 \\ -3 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} 8x + 4y \\ -3x - 2y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$8x + 4y = x \Rightarrow 7x + 4y = 0 \Rightarrow y = -\frac{7}{4}x$

$-3x - 2y = y \Rightarrow 3x + 3y = 0 \Rightarrow y = -x$

$y = -\frac{7}{4}x$ and $y = -x$ intersect only at the origin, so there are no lines of invariant points.

14)

$$\begin{pmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{3}{5}x + \frac{4}{5}y \\ \frac{4}{5}x - \frac{3}{5}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\frac{3}{5}x + \frac{4}{5}y = x \Rightarrow -\frac{2}{5}x + \frac{4}{5}y = 0 \Rightarrow -2x + 4y = 0 \Rightarrow y = \frac{1}{2}x$$

$$\frac{4}{5}x - \frac{3}{5}y = y \Rightarrow -\frac{4}{5}x + \frac{8}{5}y = 0 \Rightarrow -4x + 8y = 0 \Rightarrow y = \frac{1}{2}x$$

As the transformation is a reflection, any points that lie on the mirror line must be invariant, and hence the mirror line itself is a line of invariant points. The equation of the mirror line is $y = \frac{1}{2}x$

15)

$$\begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & \frac{k^2-1}{1+k^2} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{1-k^2}{1+k^2}x + \frac{2k}{1+k^2}y \\ \frac{2k}{1+k^2}x + \frac{k^2-1}{1+k^2}y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\frac{1-k^2}{1+k^2}x + \frac{2k}{1+k^2}y = x \Rightarrow (1-k^2)x + 2ky = (1+k^2)x$$

$$\Rightarrow x - k^2x + 2ky = x + k^2x \Rightarrow 2ky = 2k^2x \Rightarrow y = kx$$

$$\frac{2k}{1+k^2}x + \frac{k^2-1}{1+k^2}y = y \Rightarrow 2kx + (k^2-1)y = (1+k^2)y$$

$$\Rightarrow 2kx + k^2y - y = y + k^2y \Rightarrow 2kx = 2y \Rightarrow y = kx$$

$\therefore y = kx$ is a line of invariant points

Alternative Method:

$$\begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & \frac{k^2-1}{1+k^2} \end{pmatrix}\begin{pmatrix} x \\ kx \end{pmatrix} = \begin{pmatrix} \frac{1-k^2}{1+k^2}x + \frac{2k}{1+k^2} \times kx \\ \frac{2k}{1+k^2}x + \frac{k^2-1}{1+k^2} \times kx \end{pmatrix} = \begin{pmatrix} \frac{1-k^2}{1+k^2}x + \frac{2k^2}{1+k^2}x \\ \frac{2k}{1+k^2}x + \frac{k^3-k}{1+k^2}x \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1-k^2+2k^2}{1+k^2}x \\ \frac{2k+k^3-k}{1+k^2}x \end{pmatrix} = \begin{pmatrix} \frac{1+k^2}{1+k^2}x \\ \frac{k+k^3}{1+k^2}x \end{pmatrix} = \begin{pmatrix} x \\ \frac{(1+k^2)k}{1+k^2}x \end{pmatrix} = \begin{pmatrix} x \\ kx \end{pmatrix}$$

$\therefore y = kx$ is a line of invariant points

16) a)

$$\begin{pmatrix} 1 & -1 & 2 \\ -1 & 0 & 3 \\ 3 & -1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} x - y + 2z \\ -x + 3z \\ 3x - y \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$x - y + 2z = x \Rightarrow -y + 2z = 0 \Rightarrow y = 2z$ (1)

$-x + 3z = y \Rightarrow -x - y + 3z = 0$ (2)

$3x - y = z \Rightarrow 3x - y - z = 0$ (3)

Substitute (1) into (2):

$-x - (2z) + 3z = 0 \Rightarrow -x + z = 0 \Rightarrow x = z$ (4)

So $2z = y$ and $2z = 2x$

The line of invariant points has cartesian equation $2x = y = 2z$ or $\frac{x}{\frac{1}{2}} = y = \frac{z}{\frac{1}{2}}$

16) b)

$$\begin{pmatrix} -1 & 0 & 4 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} -x + 4z \\ 2x + 3y + 2z \\ x + y + 2z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$-x + 4z = x \Rightarrow -2x + 4z = 0 \Rightarrow x = 2z$ (1)

$2x + 3y + 2z = y \Rightarrow 2x + 2y + 2z = 0 \Rightarrow x + y + z = 0$ (2)

$x + y + 2z = z \Rightarrow x + y + z = 0$ (3)

Substitute (1) into (2):

$(2z) + y + z = 0 \Rightarrow y + 3z = 0 \Rightarrow y = -3z$ (4)

So $z = \frac{1}{2}x$ and $z = -\frac{1}{3}y$

The line of invariant points has cartesian equation $\frac{1}{2}x = -\frac{1}{3}y = z$ or $\frac{x}{2} = \frac{y}{-3} = z$

17) a)

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-x = x \Rightarrow x = 0$

$y = y \Rightarrow y \in \mathbb{R}$

The line $x = 0$ is a line of invariant points.

Any line of the form $y = k$, where k is a constant, is an invariant line.

17) b)

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$-y = x \Rightarrow y = -x$

$x = y \Rightarrow y = x$

These lines intersect at $(0, 0)$, and so the origin is the only invariant point.

17) c)

$\begin{pmatrix}3 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}\Rightarrow\begin{pmatrix}3x\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}$

$3x=x\Rightarrow x=0$

$y=y\Rightarrow y\in\mathbb{R}$

The line $x=0$ is a line of invariant points.

Any line of the form $y=k$, where k is a constant, is an invariant line.

17) d)

$\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}\Rightarrow\begin{pmatrix}x\\ -y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}$

$x=x\Rightarrow x\in\mathbb{R}$

$-y=y\Rightarrow y=0$

The line $y=0$ is a line of invariant points.

Any line of the form $x=k$, where k is a constant, is an invariant line.

17) e)

$\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}\Rightarrow\begin{pmatrix}-x\\ -y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}$

$-x=x\Rightarrow x=0$

$-y=y\Rightarrow y=0$

These lines intersect at $(0,0)$, and so the origin is the only invariant point.

17) f)

$\begin{pmatrix}1 & 0\\ 0 & 2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}\Rightarrow\begin{pmatrix}x\\ 2y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}$

$x=x\Rightarrow x\in\mathbb{R}$

$2y=y\Rightarrow y=0$

The line $y=0$ is a line of invariant points.

Any line of the form $x=k$, where k is a constant, is an invariant line.

17) g)

$\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}\Rightarrow\begin{pmatrix}y\\ x\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}$

$y=x$

$x=y\Rightarrow y=x$

The line $y=x$ is a line of invariant points.

Any line of the form $y=-x+k$, where k is a constant, is an invariant line.

17) h)

$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{pmatrix} -y \\ -x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$-y = x \Rightarrow y = -x$

$-x = y \Rightarrow y = -x + c$

The line $y = -x$ is a line of invariant points.

Any line of the form $y = x + k$, where k is a constant, is an invariant line.

18) a)

The x-axis is a line of invariant points.

18) b)

The plane $y = 0$ is a plane of invariant points.

Any line perpendicular to $y = 0$ would be an invariant line, i.e. $\mathbf{r} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

19) a)

$\begin{pmatrix} 3 & 4 \\ 2 & 5 \end{pmatrix}\begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \times 3 + 4 \times 3 \\ 2 \times 3 + 5 \times 3 \end{pmatrix} = \begin{pmatrix} 21 \\ 21 \end{pmatrix} = 7\begin{pmatrix} 3 \\ 3 \end{pmatrix}$

$\therefore k = 7$

19) b)

$y = x$ is an invariant line under the transformation.

20) a)

$\begin{pmatrix} -2 & 9 \\ 3 & -8 \end{pmatrix}\begin{pmatrix} -4 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \times -4 + 9 \times 4 \\ 3 \times -4 - 8 \times 4 \end{pmatrix} = \begin{pmatrix} 44 \\ -44 \end{pmatrix} = -11\begin{pmatrix} -4 \\ 4 \end{pmatrix}$

$\therefore k = -11$

20) b)

$y = -x$ is an invariant line under the transformation.

20) c)

$\begin{pmatrix} -2 & 9 \\ 3 & -8 \end{pmatrix}\begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \times 6 + 9 \times 2 \\ 3 \times 6 - 8 \times 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \end{pmatrix} = 1\begin{pmatrix} 6 \\ 2 \end{pmatrix}$

$\therefore p = 1$

20) d)

$y = \frac{2}{6}x \Rightarrow y = \frac{1}{3}x$ is a line of invariant points under the transformation, as $(6, 2)$ maps to itself.

21)

$$\begin{pmatrix}2 & 1\\ -5 & 8\end{pmatrix}\begin{pmatrix}x\\ 5x\end{pmatrix}=\begin{pmatrix}2\times x+5\times x\\ -5\times x+8\times 5x\end{pmatrix}=\begin{pmatrix}7x\\ 35x\end{pmatrix}=7\begin{pmatrix}x\\ 5x\end{pmatrix}$$

$\therefore y=5x$ is an invariant line under the transformation.

22)

$$\begin{pmatrix}-3 & 2\\ 5 & 6\end{pmatrix}\begin{pmatrix}x\\ -\frac{1}{2}x\end{pmatrix}=\begin{pmatrix}-3\times x+2\times\left(-\frac{1}{2}x\right)\\ 5\times x+6\times\left(-\frac{1}{2}x\right)\end{pmatrix}=\begin{pmatrix}-4x\\ 2x\end{pmatrix}=-4\begin{pmatrix}x\\ -\frac{1}{2}x\end{pmatrix}$$

$\therefore y=-\frac{1}{2}x$ is an invariant line under the transformation.

23)

$$\begin{pmatrix}-7 & -4\\ -4 & 9\end{pmatrix}\begin{pmatrix}x\\ \frac{1}{2-\sqrt{5}}x\end{pmatrix}=\begin{pmatrix}-7\times x-4\times\frac{1}{2-\sqrt{5}}x\\ -4\times x+9\times\frac{1}{2-\sqrt{5}}x\end{pmatrix}=\begin{pmatrix}-7x-\frac{4}{2-\sqrt{5}}x\\ -4x+\frac{9}{2-\sqrt{5}}x\end{pmatrix}=\begin{pmatrix}\frac{-7(2-\sqrt{5})-4}{2-\sqrt{5}}x\\ \frac{-4(2-\sqrt{5})+9}{2-\sqrt{5}}x\end{pmatrix}$$

$$=\begin{pmatrix}\frac{-14+7\sqrt{5}-4}{2-\sqrt{5}}x\\ \frac{-8+4\sqrt{5}+9}{2-\sqrt{5}}x\end{pmatrix}=\begin{pmatrix}\frac{-18+7\sqrt{5}}{2-\sqrt{5}}x\\ \frac{1+4\sqrt{5}}{2-\sqrt{5}}x\end{pmatrix}=\begin{pmatrix}\frac{-18+7\sqrt{5}}{2-\sqrt{5}}\times\frac{2+\sqrt{5}}{2+\sqrt{5}}x\\ \frac{1+4\sqrt{5}}{2-\sqrt{5}}x\end{pmatrix}=\begin{pmatrix}\frac{-36-18\sqrt{5}+14\sqrt{5}+35}{4-5}x\\ \frac{1+4\sqrt{5}}{2-\sqrt{5}}x\end{pmatrix}$$

$$=\begin{pmatrix}\frac{-1-4\sqrt{5}}{-1}x\\ \frac{1+4\sqrt{5}}{2-\sqrt{5}}x\end{pmatrix}=\begin{pmatrix}(1+4\sqrt{5})x\\ \frac{1+4\sqrt{5}}{2-\sqrt{5}}x\end{pmatrix}=(1+4\sqrt{5})\begin{pmatrix}x\\ \frac{1}{2-\sqrt{5}}x\end{pmatrix}$$

$\therefore y=\frac{1}{2-\sqrt{5}}x$ is an invariant line under the transformation.

24)

$$\begin{pmatrix}\mathrm{e}^4 & \mathrm{e}^{-2}\\ \mathrm{e}^2 & \mathrm{e}^{-4}\end{pmatrix}\begin{pmatrix}x\\ \frac{1}{\mathrm{e}^2}x\end{pmatrix}=\begin{pmatrix}\mathrm{e}^4\times x+\mathrm{e}^{-2}\times\left(\frac{1}{\mathrm{e}^2}x\right)\\ \mathrm{e}^2\times x+\mathrm{e}^{-4}\times\left(\frac{1}{\mathrm{e}^2}x\right)\end{pmatrix}=\begin{pmatrix}\mathrm{e}^4x+\mathrm{e}^{-4}x\\ \mathrm{e}^2x+\mathrm{e}^{-6}x\end{pmatrix}$$

$$=\begin{pmatrix}(\mathrm{e}^4+\mathrm{e}^{-4})x\\ (\mathrm{e}^2+\mathrm{e}^{-6})x\end{pmatrix}=(\mathrm{e}^4+\mathrm{e}^{-4})\begin{pmatrix}x\\ \frac{\mathrm{e}^2+\mathrm{e}^{-6}}{\mathrm{e}^4+\mathrm{e}^{-4}}x\end{pmatrix}=(\mathrm{e}^4+\mathrm{e}^{-4})\begin{pmatrix}x\\ \frac{\mathrm{e}^2+\mathrm{e}^{-6}}{\mathrm{e}^4+\mathrm{e}^{-4}}\times\frac{\mathrm{e}^2}{\mathrm{e}^2}x\end{pmatrix}$$

$$=(\mathrm{e}^4+\mathrm{e}^{-4})\begin{pmatrix}x\\ \frac{\mathrm{e}^4+\mathrm{e}^{-4}}{\mathrm{e}^4+\mathrm{e}^{-4}}\times\frac{1}{\mathrm{e}^2}x\end{pmatrix}=(\mathrm{e}^4+\mathrm{e}^{-4})\begin{pmatrix}x\\ \frac{1}{\mathrm{e}^2}x\end{pmatrix}$$

$\therefore y=\frac{1}{\mathrm{e}^2}x$ is an invariant line under the transformation.

25)

$$\begin{pmatrix}\ln 2 & \ln 8\\ \ln 4 & \ln 2\end{pmatrix}\begin{pmatrix}x\\ \frac{\sqrt{6}}{3}x\end{pmatrix}=\begin{pmatrix}\ln 2 & 3\ln 2\\ 2\ln 2 & \ln 2\end{pmatrix}\begin{pmatrix}x\\ \frac{\sqrt{6}}{3}x\end{pmatrix}=\ln 2\begin{pmatrix}1 & 3\\ 2 & 1\end{pmatrix}\begin{pmatrix}x\\ \frac{\sqrt{6}}{3}x\end{pmatrix}$$

$$=\ln 2\begin{pmatrix}1\times x+3\times\frac{\sqrt{6}}{3}x\\ 2\times x+1\times\frac{\sqrt{6}}{3}x\end{pmatrix}=\ln 2\begin{pmatrix}x+\sqrt{6}x\\ 2x+\frac{\sqrt{6}}{3}x\end{pmatrix}=\ln 2\begin{pmatrix}(1+\sqrt{6})x\\ \left(2+\frac{\sqrt{6}}{3}\right)x\end{pmatrix}$$

$$=(1+\sqrt{6})\ln 2\begin{pmatrix}x\\ \frac{2+\frac{\sqrt{6}}{3}}{1+\sqrt{6}}x\end{pmatrix}=(1+\sqrt{6})\ln 2\begin{pmatrix}x\\ \frac{6+\sqrt{6}}{3+3\sqrt{6}}x\end{pmatrix}=(1+\sqrt{6})\ln 2\begin{pmatrix}x\\ \frac{6+\sqrt{6}}{3+3\sqrt{6}}\times\frac{3-3\sqrt{6}}{3-3\sqrt{6}}x\end{pmatrix}$$

$$=(1+\sqrt{6})\ln 2\begin{pmatrix}x\\ \frac{-15\sqrt{6}}{-45}x\end{pmatrix}=(1+\sqrt{6})\ln 2\begin{pmatrix}x\\ \frac{\sqrt{6}}{3}x\end{pmatrix}$$

$\therefore y=\frac{\sqrt{6}}{3}x$ is an invariant line under the transformation.

26)

$$\begin{pmatrix}\sin k\cos k & \sin k\cos k\\ \sin k\cos k & -\sin k\cos k\end{pmatrix}\begin{pmatrix}x\\ \frac{1}{1+\sqrt{2}}x\end{pmatrix}=\begin{pmatrix}\sin k\cos k\times x+\sin k\cos k\times\frac{1}{1+\sqrt{2}}x\\ \sin k\cos k\times x+(-\sin k\cos k)\times\frac{1}{1+\sqrt{2}}x\end{pmatrix}$$

$$=\begin{pmatrix}\sin k\cos k\left(1+\frac{1}{1+\sqrt{2}}\right)x\\ \sin k\cos k\left(1-\frac{1}{1+\sqrt{2}}\right)x\end{pmatrix}=\sin k\cos k\begin{pmatrix}\left(1+\frac{1}{1+\sqrt{2}}\right)x\\ \left(1-\frac{1}{1+\sqrt{2}}\right)x\end{pmatrix}$$

$$=\sin k\cos k\left(1+\frac{1}{1+\sqrt{2}}\right)\begin{pmatrix}x\\ \frac{1-\frac{1}{1+\sqrt{2}}}{1+\frac{1}{1+\sqrt{2}}}x\end{pmatrix}$$

$$=\sin k\cos k\left(1+\frac{1}{1+\sqrt{2}}\right)\begin{pmatrix}x\\ \frac{1+\sqrt{2}-1}{1+\sqrt{2}+1}x\end{pmatrix}$$

$$=\sin k\cos k\left(1+\frac{1}{1+\sqrt{2}}\right)\begin{pmatrix}x\\ \frac{\sqrt{2}}{2+\sqrt{2}}x\end{pmatrix}$$

$$=\sin k\cos k\left(1+\frac{1}{1+\sqrt{2}}\right)\begin{pmatrix}x\\ \frac{1}{\frac{2}{\sqrt{2}}+1}x\end{pmatrix}$$

$$=\sin k\cos k\left(1+\frac{1}{1+\sqrt{2}}\right)\begin{pmatrix}x\\ \frac{1}{1+\sqrt{2}}x\end{pmatrix}$$

$\therefore y=\frac{1}{1+\sqrt{2}}x$ is an invariant line under the transformation.

27) a)

$\begin{pmatrix}1 & 1\\ -2 & 4\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}\Rightarrow\begin{pmatrix}x+y\\ -2x+4y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}$

$X=x+y\Rightarrow X=x+(mx+c)\Rightarrow X=(m+1)x+c$

$Y=-2x+4y\Rightarrow Y=-2x+4(mx+c)\Rightarrow Y=(4m-2)x+4c$

$Y=mX+c$:

$(4m-2)x+4c=m\big((m+1)x+c\big)+c$

$\Rightarrow 4mx-2x+4c=m^2x+mx+mc+c$

$\Rightarrow 0=m^2x-3mx+2x+mc-3c$

$\Rightarrow 0=(m^2-3m+2)x+(m-3)c$

Compare coefficients:

$m^2-3m+2=0\Rightarrow(m-2)(m-1)=0\Rightarrow m=2$ or $m=1$

$(m-3)c=0\Rightarrow m=3$ or $c=0$

So either $m=2$ and $c=0$, or $m=1$ and $c=0$

$\therefore y=2x$ and $y=x$ are invariant lines

$\begin{pmatrix}1 & 1\\ -2 & 4\end{pmatrix}\begin{pmatrix}1\\ 2\end{pmatrix}=\begin{pmatrix}3\\ 6\end{pmatrix}$ and $\begin{pmatrix}1 & 1\\ -2 & 4\end{pmatrix}\begin{pmatrix}1\\ 1\end{pmatrix}=\begin{pmatrix}2\\ 2\end{pmatrix}$ so neither are lines of invariant points.

27) b)

$\begin{pmatrix}6 & 2\\ -4 & 15\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}\Rightarrow\begin{pmatrix}6x+2y\\ -4x+15y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}$

$X=6x+2y\Rightarrow X=6x+2(mx+c)\Rightarrow X=(2m+6)x+2c$

$Y=-4x+15y\Rightarrow Y=-4x+15(mx+c)\Rightarrow Y=(15m-4)x+15c$

$Y=mX+c$:

$(15m-4)x+15c=m\big((2m+6)x+2c\big)+c$

$\Rightarrow 15mx-4x+15c=2m^2x+6mx+2mc+c$

$\Rightarrow 0=2m^2x-9mx+4x+2mc-14c$

$\Rightarrow 0=(2m^2-9m+4)x+(2m-14)c$

Compare coefficients:

$2m^2-9m+4=0\Rightarrow(2m-1)(m-4)=0\Rightarrow m=\frac{1}{2}$ or $m=4$

$(2m-14)c=0\Rightarrow m=7$ or $c=0$

So either $m=\frac{1}{2}$ and $c=0$, or $m=4$ and $c=0$

$\therefore y=\frac{1}{2}x$ and $y=4x$ are invariant lines

$\begin{pmatrix}6 & 2\\ -4 & 15\end{pmatrix}\begin{pmatrix}1\\ \frac{1}{2}\end{pmatrix}=\begin{pmatrix}7\\ \frac{7}{2}\end{pmatrix}$ and $\begin{pmatrix}6 & 2\\ -4 & 15\end{pmatrix}\begin{pmatrix}1\\ 4\end{pmatrix}=\begin{pmatrix}14\\ 56\end{pmatrix}$ so neither are lines of invariant points.

27) c)

$\begin{pmatrix} 5 & -3 \\ -1 & 7 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 5x-3y \\ -x+7y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$

$X = 5x - 3y \Rightarrow X = 5x - 3(mx + c) \Rightarrow X = (-3m + 5)x - 3c$

$Y = -x + 7y \Rightarrow Y = -x + 7(mx + c) \Rightarrow Y = (7m - 1)x + 7c$

$Y = mX + c$:

$(7m - 1)x + 7c = m\big((-3m + 5)x - 3c\big) + c$

$\Rightarrow 7mx - x + 7c = -3m^2x + 5mx - 3mc + c$

$\Rightarrow 3m^2x + 2mx - x + 3mc + 6c = 0$

$\Rightarrow (3m^2 + 2m - 1)x + (3m + 6)c = 0$

Compare coefficients:

$3m^2 + 2m - 1 = 0 \Rightarrow (3m - 1)(m + 1) = 0 \Rightarrow m = \frac{1}{3}$ or $m = -1$

$(3m + 6)c = 0 \Rightarrow m = -2$ or $c = 0$

So either $m = \frac{1}{3}$ and $c = 0$, or $m = -1$ and $c = 0$

$\therefore y = \frac{1}{3}x$ and $y = -x$ are invariant lines

$\begin{pmatrix} 5 & -3 \\ -1 & 7 \end{pmatrix}\begin{pmatrix} 1 \\ \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 4 \\ \frac{4}{3} \end{pmatrix}$ and $\begin{pmatrix} 5 & -3 \\ -1 & 7 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 8 \\ -8 \end{pmatrix}$ so neither are lines of invariant points.

27) d)

$\begin{pmatrix} -25 & 24 \\ 12 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} -25x+24y \\ 12x+3y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$

$X = -25x + 24y \Rightarrow X = -25x + 24(mx + c) \Rightarrow X = (24m - 25)x + 24c$

$Y = 12x + 3y \Rightarrow Y = 12x + 3(mx + c) \Rightarrow Y = (3m + 12)x + 3c$

$Y = mX + c$:

$(3m + 12)x + 3c = m\big((24m - 25)x + 24c\big) + c$

$\Rightarrow 3mx + 12x + 3c = 24m^2x - 25mx + 24mc + c$

$\Rightarrow 0 = 24m^2x - 28mx - 12x + 24mc - 2c$

$\Rightarrow 0 = 12m^2x - 14mx - 6x + 12mc - c$

$\Rightarrow 2(6m^2 - 7m - 3)x + (12m - 1)c = 0$

Compare coefficients:

$6m^2 - 7m - 3 = 0 \Rightarrow (3m + 1)(2m - 3) = 0 \Rightarrow m = -\frac{1}{3}$ or $m = \frac{3}{2}$

$(12m - 1)c = 0 \Rightarrow m = \frac{1}{12}$ or $c = 0$

So either $m = -\frac{1}{3}$ and $c = 0$, or $m = \frac{3}{2}$ and $c = 0$

$\therefore y = -\frac{1}{3}x$ and $y = \frac{3}{2}x$ are invariant lines

$\begin{pmatrix} -25 & 24 \\ 12 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} -33 \\ 11 \end{pmatrix}$ and $\begin{pmatrix} -25 & 24 \\ 12 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 11 \\ \frac{33}{2} \end{pmatrix}$ so neither are lines of invariant points.

27) e)

$$\begin{pmatrix} \frac{3}{7} & -\frac{24}{7} \\ -\frac{4}{7} & -\frac{17}{7} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{3}{7}x - \frac{24}{7}y \\ -\frac{4}{7}x - \frac{17}{7}y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$X = \frac{3}{7}x - \frac{24}{7}y \Rightarrow X = \frac{3}{7}x - \frac{24}{7}(mx + c) \Rightarrow X = \left(-\frac{24}{7}m + \frac{3}{7}\right)x - \frac{24}{7}c$

$Y = -\frac{4}{7}x - \frac{17}{7}y \Rightarrow Y = -\frac{4}{7}x - \frac{17}{7}(mx + c) \Rightarrow Y = \left(-\frac{17}{7}m - \frac{4}{7}\right)x - \frac{17}{7}c$

$Y = mX + c$:

$\left(-\frac{17}{7}m - \frac{4}{7}\right)x - \frac{17}{7}c = m\left(\left(-\frac{24}{7}m + \frac{3}{7}\right)x - \frac{24}{7}c\right) + c$

$\Rightarrow -\frac{17}{7}mx - \frac{4}{7}x - \frac{17}{7}c = -\frac{24}{7}m^2x + \frac{3}{7}mx - \frac{24}{7}mc + c$

$\Rightarrow -17mx - 4x - 17c = -24m^2x + 3mx - 24mc + 7c$

$\Rightarrow 24m^2x - 20mx - 4x + 24mc - 24c = 0$

$\Rightarrow 6m^2x - 5mx - x + 6mc - 6c = 0$

$\Rightarrow (6m^2 - 5m - 1)x + (6m - 6)c = 0$

Compare coefficients:

$6m^2 - 5m - 1 = 0 \Rightarrow (6m + 1)(m - 1) = 0 \Rightarrow m = -\frac{1}{6}$ or $m = 1$

$(6m - 6)c = 0 \Rightarrow m = 1$ or $c = 0$

So either $m = 1$ and c is any real value, or $m = -\frac{1}{6}$ and $c = 0$

$\therefore y = x + c$ and $y = -\frac{1}{6}x$ are invariant lines

$\begin{pmatrix} \frac{3}{7} & -\frac{24}{7} \\ -\frac{4}{7} & -\frac{17}{7} \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$ so $y = x + c$ is an invariant line,

but $\begin{pmatrix} \frac{3}{7} & -\frac{24}{7} \\ -\frac{4}{7} & -\frac{17}{7} \end{pmatrix}\begin{pmatrix} 1 \\ -\frac{1}{6} \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{1}{6} \end{pmatrix}$ so $y = -\frac{1}{6}x$ is a line of invariant points.

27) f)

$$\begin{pmatrix} \frac{39}{19} & -\frac{5}{19} \\ \frac{4}{19} & \frac{18}{19} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{39}{19}x - \frac{5}{19}y \\ \frac{4}{19}x + \frac{18}{19}y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$X = \frac{39}{19}x - \frac{5}{19}y \Rightarrow X = \frac{39}{19}x - \frac{5}{19}(mx + c) \Rightarrow X = \left(-\frac{5}{19}m + \frac{39}{19}\right)x - \frac{5}{19}c$

$Y = \frac{4}{19}x + \frac{18}{19}y \Rightarrow Y = \frac{4}{19}x + \frac{18}{19}(mx + c) \Rightarrow Y = \left(\frac{18}{19}m + \frac{4}{19}\right)x + \frac{18}{19}c$

$Y = mX + c$:

$\left(\frac{18}{19}m + \frac{4}{19}\right)x + \frac{18}{19}c = m\left(\left(-\frac{5}{19}m + \frac{39}{19}\right)x - \frac{5}{19}c\right) + c$

$\Rightarrow \frac{18}{19}mx + \frac{4}{19}x + \frac{18}{19}c = -\frac{5}{19}m^2x + \frac{39}{19}mx - \frac{5}{19}mc + c$

$\Rightarrow 18mx + 4x + 18c = -5m^2x + 39mx - 5mc + 19c$

$\Rightarrow 5m^2x - 21mx + 4x + 5mc - c = 0$

$\Rightarrow (5m^2 - 21m + 4)x + (5m - 1)c = 0$

Compare coefficients:

$5m^2 - 21m + 4 = 0 \Rightarrow (5m - 1)(m - 4) = 0 \Rightarrow m = \frac{1}{5}$ or $m = 4$

$(5m - 1)c = 0 \Rightarrow m = \frac{1}{5}$ or $c = 0$

So either $m = \frac{1}{5}$ and c is any real value, or $m = 4$ and $c = 0$

$\therefore y = \frac{1}{5}x + c$ and $y = 4x$ are invariant lines

$\begin{pmatrix} \frac{39}{19} & -\frac{5}{19} \\ \frac{4}{19} & \frac{18}{19} \end{pmatrix}\begin{pmatrix} 1 \\ \frac{1}{5} \end{pmatrix} = \begin{pmatrix} 2 \\ \frac{2}{5} \end{pmatrix}$ so $y = \frac{1}{5}x + c$ is an invariant line,

but $\begin{pmatrix} \frac{39}{19} & -\frac{5}{19} \\ \frac{4}{19} & \frac{18}{19} \end{pmatrix}\begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ so $y = 4x$ is a line of invariant points.

28)

$\begin{pmatrix}2 & 0\\ 1 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}\Rightarrow\begin{pmatrix}2x\\ x+y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}$

$X=2x$

$Y=x+y\Rightarrow Y=x+(mx+c)\Rightarrow Y=(m+1)x+c$

$Y=mX+c$:

$(m+1)x+c=m(2x)+c$

$\Rightarrow mx+x+c=2mx+c$

$\Rightarrow 0=mx-x$

$\Rightarrow 0=(m-1)x$

Compare coefficients:

$m-1=0\Rightarrow m=1$

$x=0$

So $m=1$ and c can be any real value, or $x=0$

$\therefore y=x+c$ and $x=0$ are invariant lines

$\begin{pmatrix}2 & 0\\ 1 & 1\end{pmatrix}\begin{pmatrix}1\\ 1\end{pmatrix}=\begin{pmatrix}2\\ 2\end{pmatrix}$ so $y=x+c$ is an invariant line, but $\begin{pmatrix}2 & 0\\ 1 & 1\end{pmatrix}\begin{pmatrix}0\\ 1\end{pmatrix}=\begin{pmatrix}0\\ 1\end{pmatrix}$ so $x=0$ is a line of invariant points.

29)

$\begin{pmatrix}3 & 1\\ 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}\Rightarrow\begin{pmatrix}3x+y\\ y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}$

$X=3x+y\Rightarrow X=3x+(mx+c)\Rightarrow X=(3+m)x+c$

$Y=y\Rightarrow Y=mx+c$

$Y=mX+c$:

$mx+c=m\big((3+m)x+c\big)+c$

$\Rightarrow mx+c=3mx+m^2x+mc+c$

$\Rightarrow 0=2mx+m^2x+mc$

$\Rightarrow 0=(2m+m^2)x+mc$

Compare coefficients:

$2m+m^2=0\Rightarrow m(2+m)=0\Rightarrow m=0$ or $m=-2$

$mc=0\Rightarrow m=0$ or $c=0$

So $m=0$ and c can be any real value

Or $m=-2$ and $c=0$

$\therefore y=c$ and $y=-2x$ are invariant lines

$\begin{pmatrix}3 & 1\\ 0 & 1\end{pmatrix}\begin{pmatrix}0\\ 1\end{pmatrix}=\begin{pmatrix}1\\ 1\end{pmatrix}$ so $y=c$ is an invariant line, but $\begin{pmatrix}3 & 1\\ 0 & 1\end{pmatrix}\begin{pmatrix}1\\ -2\end{pmatrix}=\begin{pmatrix}1\\ -2\end{pmatrix}$ so $y=-2x$ is a line of invariant points.

30)

$\begin{pmatrix} 2 & -\frac{1}{2} \\ 8 & -3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 2x - \frac{1}{2}y \\ 8x - 3y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$

$X = 2x - \frac{1}{2}y \Rightarrow X = 2x - \frac{1}{2}(mx + c) \Rightarrow X = \left(-\frac{1}{2}m + 2\right)x - \frac{1}{2}c$

$Y = 8x - 3y \Rightarrow Y = 8x - 3(mx + c) \Rightarrow Y = (-3m + 8)x - 3c$

$Y = mX + c$:

$(-3m + 8)x - 3c = m\left(\left(-\frac{1}{2}m + 2\right)x - \frac{1}{2}c\right) + c$

$\Rightarrow -3mx + 8x - 3c = -\frac{1}{2}m^2x + 2mx - \frac{1}{2}mc + c$

$\Rightarrow \frac{1}{2}m^2x - 5mx + 8x + \frac{1}{2}mc - 4c = 0$

$\Rightarrow m^2x - 10mx + 16x + mc - 8c = 0$

$\Rightarrow (m^2 - 10m + 16)x + (m - 8)c = 0$

Compare coefficients:

$m^2 - 10m + 16 = 0 \Rightarrow (m - 8)(m - 2) = 0 \Rightarrow m = 8$ or $m = 2$

$(m - 8)c = 0 \Rightarrow m = 8$ or $c = 0$

So either $m = 8$ and c can be any real value, or $m = 2$ and $c = 0$

$\therefore y = 8x + c$ and $y = 2x$ are invariant lines

$\begin{pmatrix} 2 & -\frac{1}{2} \\ 8 & -3 \end{pmatrix}\begin{pmatrix} 1 \\ 8 \end{pmatrix} = \begin{pmatrix} -2 \\ -16 \end{pmatrix}$ so $y = 8x + c$ is an invariant line, but $\begin{pmatrix} 2 & -\frac{1}{2} \\ 8 & -3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ so $y = 2x$ is a line of invariant points.

So $p = 2$ and $q = 8$

31)

$\begin{pmatrix} -6 & -3 \\ 5 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} -6x - 3y \\ 5x - y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$

$X = -6x - 3y \Rightarrow X = -6x - 3(mx + c) \Rightarrow X = (-3m - 6)x - 3c$

$Y = 5x - y \Rightarrow Y = 5x - (mx + c) \Rightarrow Y = (-m + 5)x - c$

$Y = mX + c$:

$(-m + 5)x - c = m\big((-3m - 6)x - 3c\big) + c$

$\Rightarrow -mx + 5x - c = -3m^2x - 6mx - 3mc + c$

$\Rightarrow 3m^2x + 5mx + 5x + 3mc - 2c = 0$

$\Rightarrow (3m^2 + 5m + 5)x + (3m - 2)c = 0$

Compare coefficients:

$3m^2 + 5m + 5 = 0$

$b^2 - 4ac = (5)^2 - 4(3)(5) = -35 < 0$ so no real roots

$(3m - 2)c = 0 \Rightarrow m = \frac{2}{3}$ or $c = 0$

So there are no invariant lines

32)

$\begin{pmatrix}-10 & 6\\ -8 & -15\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}\Rightarrow\begin{pmatrix}-10x+6y\\ -8x-15y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}$

$X=-10x+6y\Rightarrow X=-10x+6(mx+c)\Rightarrow X=(6m-10)x+6c$

$Y=-8x-15y\Rightarrow Y=-8x-15(mx+c)\Rightarrow Y=(-15m-8)x-15c$

$Y=mX+c$:

$(-15m-8)x-15c=m\big((6m-10)x+6c\big)+c$

$\Rightarrow -15mx-8x-15c=6m^2x-10mx+6mc+c$

$\Rightarrow 0=6m^2x+5mx+8x+6mc+16c$

$\Rightarrow 0=(6m^2+5m+8)x+(6m+16)c$

Compare coefficients:

$6m^2+5m+8=0$

$b^2-4ac=(5)^2-4(6)(8)=-167<0$ so no real roots

$(6m+16)c=0\Rightarrow m=-\frac{16}{6}=-\frac{8}{3}$ or $c=0$

So there are no invariant lines

33) a)

$\begin{pmatrix}\frac{24}{25} & \frac{7}{25}\\ \frac{7}{25} & -\frac{24}{25}\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}\Rightarrow\begin{pmatrix}\frac{24}{25}x+\frac{7}{25}y\\ \frac{7}{25}x-\frac{24}{25}y\end{pmatrix}=\begin{pmatrix}x\\ y\end{pmatrix}$

$\frac{24}{25}x+\frac{7}{25}y=x\Rightarrow-\frac{1}{25}x+\frac{7}{25}y=0\Rightarrow -x+7y=0\Rightarrow y=\frac{1}{7}x$

$\frac{7}{25}x-\frac{24}{25}y=y\Rightarrow\frac{7}{25}x-\frac{49}{25}y=0\Rightarrow 7x-49y=0\Rightarrow y=\frac{1}{7}x$

$\therefore y=\frac{1}{7}x$ is a line of invariant points

33) b)

$k = -7$ as the line must be perpendicular to the mirror line. The full worked solution is shown here:

$$\begin{pmatrix} \frac{24}{25} & \frac{7}{25} \\ \frac{7}{25} & -\frac{24}{25} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{24}{25}x + \frac{7}{25}y \\ \frac{7}{25}x - \frac{24}{25}y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$X = \frac{24}{25}x + \frac{7}{25}y \Rightarrow X = \frac{24}{25}x + \frac{7}{25}(mx + c) \Rightarrow X = \left(\frac{7}{25}m + \frac{24}{25}\right)x + \frac{7}{25}c$$

$$Y = \frac{7}{25}x - \frac{24}{25}y \Rightarrow Y = \frac{7}{25}x - \frac{24}{25}(mx + c) \Rightarrow Y = \left(-\frac{24}{25}m + \frac{7}{25}\right)x - \frac{24}{25}c$$

$Y = mX + c$:

$$\left(-\frac{24}{25}m + \frac{7}{25}\right)x - \frac{24}{25}c = m\left(\left(\frac{7}{25}m + \frac{24}{25}\right)x + \frac{7}{25}c\right) + c$$

$$\Rightarrow -\frac{24}{25}mx + \frac{7}{25}x - \frac{24}{25}c = \frac{7}{25}m^2x + \frac{24}{25}mx + \frac{7}{25}mc + c$$

$$\Rightarrow -24mx + 7x - 24c = 7m^2x + 24mx + 7mc + 25c$$

$$\Rightarrow 0 = 7m^2x + 48mx - 7x + 7mc + 49c$$

$$\Rightarrow 0 = (7m^2 + 48m - 7)x + (7m + 49)c$$

Compare coefficients:

$7m^2 + 48m - 7 = 0 \Rightarrow (7m - 1)(m + 7) = 0 \Rightarrow m = \frac{1}{7}$ or $m = -7$

$(7m + 49)c = 0 \Rightarrow m = -7$ or $c = 0$

So either $m = -7$ and c can be any real value, or $m = \frac{1}{7}$ and $c = 0$

$\therefore y = -7x + c$ and $y = \frac{1}{7}x$ are invariant lines

$\begin{pmatrix} \frac{24}{25} & \frac{7}{25} \\ \frac{7}{25} & -\frac{24}{25} \end{pmatrix}\begin{pmatrix} 1 \\ -7 \end{pmatrix} = \begin{pmatrix} -1 \\ 7 \end{pmatrix}$ so $y = -7x + c$ is an invariant line,

but $\begin{pmatrix} \frac{24}{25} & \frac{7}{25} \\ \frac{7}{25} & -\frac{24}{25} \end{pmatrix}\begin{pmatrix} 1 \\ \frac{1}{7} \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1}{7} \end{pmatrix}$ so $y = \frac{1}{7}x$ is a line of invariant points.

34) a)

$\begin{pmatrix}28 & -9\\ 4 & 16\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}\Rightarrow\begin{pmatrix}28x-9y\\ 4x+16y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}$

$X=28x-9y\Rightarrow X=28x-9(mx+c)\Rightarrow X=(-9m+28)x-9c$

$Y=4x+16y\Rightarrow Y=4x+16(mx+c)\Rightarrow Y=(16m+4)x+16c$

$Y=mX+c$:

$(16m+4)x+16c=m\big((-9m+28)x-9c\big)+c$

$\Rightarrow 16mx+4x+16c=-9m^2x+28mx-9mc+c$

$\Rightarrow 9m^2x-12mx+4x+9mc+15c=0$

$\Rightarrow 0=(9m^2-12m+4)x+(9m+15)c$

Compare coefficients:

$9m^2-12m+4=0\Rightarrow(3m-2)^2=0\Rightarrow m=\frac{2}{3}$

$(9m+15)c=0\Rightarrow m=-\frac{15}{9}=-\frac{5}{3}$ or $c=0$

So $m=\frac{2}{3}$ and $c=0$

$\therefore y=\frac{2}{3}x$ is an invariant line, i.e. $k=\frac{2}{3}$

34) b)

$\begin{pmatrix}28 & -9\\ 4 & 16\end{pmatrix}\begin{pmatrix}1\\ \frac{2}{3}\end{pmatrix}=\begin{pmatrix}22\\ \frac{44}{3}\end{pmatrix}$ so $y=\frac{2}{3}x$ is not a line of invariant points.

35)

$\begin{pmatrix}241 & -60\\ 980 & -244\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}\Rightarrow\begin{pmatrix}241x-60y\\ 980x-244y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}$

$X=241x-60y\Rightarrow X=241x-60(mx+c)\Rightarrow X=(-60m+241)x-60c$

$Y=980x-244y\Rightarrow Y=980x-244(mx+c)\Rightarrow Y=(-244m+980)x-244c$

$Y=mX+c$:

$(-244m+980)x-244c=m\big((-60m+241)x-60c\big)+c$

$\Rightarrow -244mx+980x-244c=-60m^2x+241mx-60mc+c$

$\Rightarrow 60m^2x-485mx+980x+60mc-245c=0$

$\Rightarrow 12m^2x-97mx+196x+12mc-49c=0$

$\Rightarrow 0=(12m^2-97m+196)x+(12m-49)c$

Compare coefficients:

$12m^2-97m+196=0\Rightarrow(12m-49)(m-4)=0\Rightarrow m=\frac{49}{12}$ or $m=4$

$(12m-49)c=0\Rightarrow m=\frac{49}{12}$ or $c=0$

So $m=\frac{49}{12}$ and $c\in\mathbb{R}$, or $m=4$ and $c=0$

$\therefore y=\frac{49}{12}x+c$ and $y=4x$ are invariant lines.

$\begin{pmatrix}241 & -60\\ 980 & -244\end{pmatrix}\begin{pmatrix}1\\ \frac{49}{12}\end{pmatrix}=\begin{pmatrix}-4\\ -\frac{49}{3}\end{pmatrix}$ so $y=\frac{49}{12}x+c$ are not lines of invariant points.

$\begin{pmatrix}241 & -60\\ 980 & -244\end{pmatrix}\begin{pmatrix}1\\ 4\end{pmatrix}=\begin{pmatrix}1\\ 4\end{pmatrix}$ so $y=4x$ is a line of invariant points.

36)

$\begin{pmatrix}2 & k\\ -3 & 4\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}\Rightarrow\begin{pmatrix}2x+ky\\ -3x+4y\end{pmatrix}=\begin{pmatrix}X\\ Y\end{pmatrix}$

$X=2x+ky\Rightarrow X=2x+k(mx)\Rightarrow X=(km+2)x$

$Y=-3x+4y\Rightarrow Y=-3x+4(mx)\Rightarrow Y=(4m-3)x$

$Y=mX+c$:

$(4m-3)x=m\big((km+2)x\big)$

$\Rightarrow 4mx-3x=km^2x+2mx$

$\Rightarrow 0=km^2x-2mx+3x$

$\Rightarrow 0=(km^2-2m+3)x$

Compare coefficients:

$km^2-2m+3=0$

No invariant lines $\Rightarrow b^2-4ac<0$

$\therefore(-2)^2-4(k)(3)<0\Rightarrow 4-12k<0\Rightarrow k>\frac{1}{3}$

So if $k>\frac{1}{3}$ then $\mathbf{A}$ has no invariant lines that pass through the origin

37)

$\begin{pmatrix} 2k & 3 \\ -1 & 2k-3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix} \Rightarrow \begin{pmatrix} 2kx+3y \\ -x+(2k-3)y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$

$X = 2kx + 3y \Rightarrow X = 2kx + 3(mx) \Rightarrow X = (2k+3m)x$

$Y = -x + (2k-3)y \Rightarrow Y = -x + (2k-3)(mx) \Rightarrow Y = (2km-3m-1)x$

$Y = mX$:

$(2km-3m-1)x = m\big((2k+3m)x\big)$

$\Rightarrow 2kmx - 3mx - x = 2kmx + 3m^2x$

$\Rightarrow 0 = 3m + x + 3m^2x$

$\Rightarrow 0 = (3m^2 + 3m + 1)x$

Compare coefficients:

$3m^2 + 3m + 1 = 0$

No invariant lines $\Rightarrow b^2 - 4ac < 0$

$\therefore 3^2 - 4(3)(1) = -3 < 0$

So there are no real roots, and hence there are no invariant lines.